

Continuous Integration Report

Group Number: Cohort 1, Group 11

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This document outlines our approach to and the methods used in the Continuous Integration (CI) for our project, and a brief account of how we have set up the CI infrastructure.

Methods & Approach

Our first priority with continuous integration was to ensure that no branch got too far ahead of the main branch. This involved making sure that all changes were merged into the main branch before each meeting on a Monday and Thursday so that we could discuss the changes in the meeting and resolve any conflicts. We also made use of Discord and Github discussions to ensure we could remain in contact remotely.

We also wanted to make sure that as much of the process was automated as possible. This helped to speed everything up which was useful for meeting the deadline. This was done by automating the build using Github actions meaning each commit produced a JAR file and ran the automated tests. In order to ensure the process ran quickly, we only executed the unit and integration tests each time and left the end-to-end tests for large, milestone commits.

The next thing was to ensure compatibility by testing the code on a designated integration machine that didn't have any of the build files, just the executable. This made sure that the game would work on a hypothetical customer machine, not just our own.

Finally, we made it a priority to fix any issues with the build as soon as possible. This meant that the issues didn't have time to grow and therefore were easier to fix meaning the final product could be delivered on time.

CI Infrastructure

We created two separate YAML files that would automate the continuous integration process. The first file defined the process to be run when a pull request was made which was to run all the automated tests and produce a test coverage report. This was to be run on the clean version of Ubuntu provided by Github actions to ensure compatibility. It was important to run these to ensure that any easy mistakes were caught before the manual review of the code which made the whole process run faster. Because there might be multiple pull requests before the actual merge into the main branch, it was important to keep this process as quick as possible which is why more steps weren't added.

We kept the push-to-main YAML file separate for clarity however it still produces a test and test coverage report. What differentiates it is that it also produces a JAR file because we wanted it to be easy to download the latest version of the game without having to pull from main and then build the project. Because this process would be run less frequently and would be more impactful on the rest of the project, we also added some more intensive tests that would only be run in this process as they would be too time consuming to be run each time a pull request was made.